

CURRICULUM VITAE

1. Name, position, academic department

Alexander Goncharov, Professor, Dept. of Mathematics, Yale University.

2. Education

M.S. (Mathematics), Moscow State University, 1982.

Ph. D. (Mathematics), Moscow and Institute of Mathematics, Novosibirsk, 1987; Scientific adviser: I.M.Gelfand.

3. Professional appointments

Professor of Mathematics, Yale University (July 2010 –)

Gretchen and Barry Mazur Chair at IHES (January 2019–December 2023)

Simons Fellow, Fall 2019, Fall 2023.

Research Professor (MSRI, Fall 2019, Spring 2023)

Member, Institute for Advanced Studies, Princeton, Fall 2017.

Distinguished Visiting Research Chair at the Perimeter Institute (Waterloo, Canada) (2014-2016)

Professor of Mathematics, Brown University (Fall 1999 - June 2010)

Senior Member, MSRI (Fall 2001)

Associate Professor of Mathematics, Brown University (1998 - 1999)

Visiting Scholar, MPI (Bonn) (Jan. - Aug. 1996, 1997)

Associate Professor of Mathematics, MIT (Fall 1995 - 1997)

Lecturer, MIT (1993 - Spring 1995)

Senior Member, MSRI (September 1992 - May 1993)

Visiting Scholar, M.I.T. (Spring 1992, Fall 1990)

Harvard Prize Fellow, Harvard University (Fall 1991)

Researcher, USSR Academy of Sciences, Moscow (1985 - 1992).

4. Academic honors

European Mathematical Society Prize at the First European Congress of Mathematicians (1992).

30-minute invited talk “Reconstruction of objects from their projections in unknown directions,” at the 12th International Congress on Electron Microscopy in Seattle (1990).

45-minute invited talk “Polylogarithms in arithmetic and geometry” at the International Congress of Mathematicians in Zürich (August 1994)

50-minute invited talk “Multiple ζ -values, Galois groups and geometry of modular varieties” at the Third European Congress of Mathematicians in Barcelona (July 2000)

An hour invited Plenary talk ”Quantum Geometry of Moduli spaces” Pekin, China, July 2024.

Talks at the Arbeitstagung, Bonn - 1997, 1999, 2011, 2017.

Gold medal on 18-th International Mathematical Olympiad (1976).

5. Service

To the University

1. A member of the Hiring Committees at the Math Department for the 2021-2025 years.
2. Math Department Representative at JBPO from December 2025.
3. Advising 26 undergraduate Math majors at Yale.

To the profession

1. An editor of the *Selecta Mathematica* journal.
2. An editor of the *Journal of Noncommutative Geometry*.

6. Research Grant

NSF grant DMS-2153059, 2022-2026.

7. Recent Invited Lectures 2024.

1. An invited lecture "Cluster unitary double quantization" at the conference *Representation Theory and Algebraic Geometry*, dedicated to 80th birthday of Joseph Bernstein, May 2025.
2. An invited talk "Cluster Lagrangians" at the conference *Perfectly matched perspectives on statistical mechanics, combinatorics and geometry*, dedicated to R. Kenyon, Marceille, June 2025.
3. An invited talk "Non-commutative Cluster Lagrangians" at the workshop *Higher Teichmuller Theory and spectral networks*, Paris, Orsay, July 8 2025.
4. A series of invited lectures "Cluster quantization and Liouville theory" at the school *Intersections of topological recursion conformal field theory and random geometry* Swiss MAP Research Station, Les Diablerets, Switzerland at August 2025.

8. Completed Research

Refereed journal articles.

1. Goncharov, A., "Infinitesimal structures connected with Hermitian symmetric spaces," *Funct. Anal. Appl.*, **15**, n. 3, 83-84, 1981.
2. Goncharov, A., "A construction of Weil representations for some simple Lie algebras," *Funct. Anal. Appl.*, **16**, n. 3, 70-71, 1982.
3. Goncharov, A., "Generalized conformal structures," *Group Theory and Homological Algebra*, Yaroslavl State Univ. Press, 99-111, 1983.
4. Vainstein, B.K. and Goncharov, A.B. "Determining the spatial orientation of arbitrarily arranged particles given their projections," *Dokl. Acad. Sci., USSR* **287**, n. 5, 1131-1134, 1986. English translation: Soviet Physics Doklady, Vol. 31, p.278.
5. Vainstein, B.K. and Goncharov, A.B. "Determining the spatial orientation of arbitrarily arranged particles given their projections," In: Proc. XIth International Congress on Electron Microscopy, Kyoto, Imura T, Mause S, Suzuki T. (eds) Japan Soc Electron Microscopy, Tokyo. pp 459-460 (1986).
6. Gelfand, I.M. and Goncharov, A. "On a characterization of Grassmannians," *Sov. Math. Dokl.* **34**, 189-193, 1987.
7. Gelfand, I.M. and Goncharov, A. "Reconstruction of a compactly supported function by its integrals over the lines intersecting a curve in the space" *Sov. Math. Dokl.* **34**, 373-376, 1987.
8. Goncharov A.B., Vainshtein B.K., Ryskin A.I., Vagin, A.I.: SOV. Physics. - Crystallography. 32, 504 (1987)
9. Goncharov, A., "Generalized conformal structures on manifolds," *Selecta Mathematica Sovietica*, **6**, n. 4, 306-340, 1987.
10. Goncharov, A. B., "Methods of integral geometry and determination of mutual orientations of identical particles located in the plane from their projections onto a line" *Dokl. Acad. Sci., USSR*, **293**, n. 2, 355-358. 1987. English translation: Soviet Physics Doklady, Vol. 32.
11. Goncharov, A., "Integral geometry and 3-Dimensional reconstruction of randomly oriented particles from their electron micrographs," *Acta Applicandae Mathematicae*, **11**, 199-211, 1988.
12. Goncharov, A., "Methods of integral geometry and recovering a compactly supported function from its projections in unknown directions," *Acta Applicandae Mathematicae*, **11**, 213-222, 1988.
13. Goncharov, A., "Admissible families of k-dimensional submanifolds," *Soviet Math. Dokl.*, **300**, n. 3, 535-539, 1988.
14. Goncharov, A., "Integral geometry on surfaces in the space," *Geometry and Physics*, **4**, 571-594, 1989.
15. Goncharov, A., "Integral geometry on k-dimensional submanifolds," *Funct. Anal. Appl.*, **23**, n. 3, 11-24, 1989.

16. Gelfand M.S., Goncharov, A., "Spatial rotational alignment of identical particles given their projections: theory and practice," *Mathematical Problems of Tomography*, 97 - 122, Math. Monographs, 81, AMS, Providence, RI, 1990.
17. Goncharov A.B., Gelfand M.S., *Determination of mutual orientation of identical particles from their projections by the moment method*. *Ultramicroscopy* 25 (1988) 317-327.
18. Gelfand M.S., Goncharov A.B.: *Spatial rotational alignment of identical particles in the case of (almost) coaxial projections*. *Ultramicroscopy*. 1989 Apr;27(3):301-6.
19. Goncharov, A., "Three-dimensional reconstruction of arbitrarily arranged identical particles given their projections," *Mathematical Problems of Tomography*, I.M.Gelfand, S.G.Gindikin, editors; in the Book: *Translations of Mathematical monographs*, **81**, American Mathematical Society, 67-96, 1990.
20. Goncharov, A., "Weil representation and hypergeometrical functions connected with Lagrangian Grassmannian," *Soviet Math. Dokl.*, **341**, n. 3, 395-399, 1990.
21. Beilinson, A., Goncharov, A., Schechtman, V. and Varchenko, A., "Aomoto dilogarithm, mixed Hodge structures and motivic cohomology of pairs of triangles on the plane," in the Book: *The Grothendieck Festschrift*, Part of Series: *Progress in Mathematics*, **86**, Birkhauser, Boston, 131-172, 1990.
22. Beilinson, A., Goncharov, A., Schechtman, V. and Varchenko, A., "Projective geometry and algebraic K-theory," *Algebra and Analysis*, **6** n. 3, 78-134, 1990.
23. Goncharov, A., "Integral geometry and subvarieties of minimal possible degree in CP^n ," *Funct. Anal. Appl.*, **24**, n. 1, 5-20, 1990.
24. Goncharov, A., "The classical trilogarithm, algebraic K-theory of fields and Dedekind zeta functions," *Bull. of AMS*, **24**, n. 1, 155-162, 1991.
25. Goncharov, A. "The classical polylogarithms, algebraic K-theory and $\zeta_F(n)$," *Proc. of the Gelfand Seminar*, Birkhauser, 113-135, 1993.
26. Goncharov, A., "When a family of submanifolds is diffeomorphic to a family of planes," *Duke Math. Journal*, **71**, n. 1, 317-331, 1993.
27. Goncharov, A., "Explicit construction of characteristic classes," *Advances in Soviet Mathematics*, **16**, Pt. 1, (special volume dedicated to I.M.Gelfand's 80th birthday), 169 - 210, 1993.
28. Goncharov, A. "Admissible double bundles," In the Book: *'75 years of the Radon transform'*, Plenum Press, 1994.
29. Goncharov, A. "Polylogarithms and Motivic Galois groups," *Proceedings of the AMS Research Summer Conference "Motives"*, *Symposium in Pure Mathematics*, **55**, 43 - 96, 1994.
30. Goncharov, A., "Geometry of configurations, polylogarithms and motivic cohomology," *Advances in Mathematics*, 1995, Vol. 114, No. 2, 1995, pp. 197-318.
31. Goncharov, A., "Chow polylogarithms and regulators," *Mathematical Research Letters*, **2**, n. 1, 95-113. 1995.

32. Goncharov, A. "Integral geometry and D-modules". Mathematical Research Letters, 1995, vol 2, N 2, 415-435.
33. Goncharov, A. "Polylogarithms in arithmetic and geometry," *Proc. of Intern. Congr. of Mathematicians*, (Zürich, 1994), 1995, vol 1, 374-387.
34. Goncharov, A. "Deninger's conjecture on L -function of elliptic curve at $s = 3$ ". Special volume dedicated to Manin's 60-th birthday (1997). 37 pages.
35. Goncharov, A. "Differential equations and Integral geometry". Advances in Math., vol. 131, N2, (1997).
36. Goncharov, A., Levin A. "Zagier's conjecture on $L(E, 2)$ ". Inventiones Math., vol 132, (1998), 393-432.
37. Goncharov, A. "The double logarithm and Manin's complex for modular curves" Mathematical Research Letters, 1997, vol. 4, No 1, 1-20.
38. Goncharov, A. "Mixed elliptic motives" Galois representations in Arithmetic Algebraic geometry, (Durham 1996), London Math. Soc. Lecture Note Ser., vol. 254, Cambridge University Press, 1998. 147-221.
39. Goncharov, A., "Multiple polylogarithms, cyclotomy and modular complexes" Mathematical Research Letters, 1998, vol. 5, No 3, 497-516.
40. Goncharov, A., "Volumes of hyperbolic manifolds and mixed Tate motives", Journal of Amer. Math. Soc. vol. 12, (1999), N2, 569-618.
41. Goncharov, A. "Geometry of the trilogarithm and the motivic Lie algebra of a field". Regulators in analysis, geometry and number theory, 127-165, Progr. Math., 171, Birkhauser Boston, Boston, MA, 2000
42. Goncharov, A. Zhao, J. "Grassmannian trilogarithms" Compositio Math. 127, (2001) N1, pp. 83-108.
43. Goncharov, A. "Explicit regulator maps on the polylogarithmic motivic complexes" In: Motives, Polylogarithms and Hodge theory. Proceedings of the Irvine conference on Polylogarithms, International Press Lecture Series 3, (2002). p. 245-277.
44. Goncharov, A. "The dihedral Lie algebras and Galois symmetries of $\pi_1^l(P^1 \setminus \{0, \infty\} \cup \mu_N)$ ". Duke Math. J., vol 100, N3, (2001), pp. 397-487. math.AG/0009121.
45. Goncharov, A. "Multiple ζ -values, Galois groups, and geometry of modular varieties". European Congress of Mathematics, Vol. I (Barcelona, 2000), 361-392, Progr. Math., 201, Birkhauser, Basel. math.AG/0204102.
46. Goncharov, A., Manin Yu. I. "Multiple zeta-motives and moduli spaces $M_{0,n}$ " Compositio Mathematica, 140 (2004), 1-14. math.AG/0204102.
47. Goncharov A. "Euclidean scissor congruence groups and mixed Tate motives over dual numbers" Math. Res. Letters, 11 (2004), vol. 6. 10001-10013. ArXiv: math.AG/0401354,
48. Goncharov, A. "Galois symmetries of fundamental groupoids and noncommutative geometry" Duke Math. J. 128 (2005), no. 2, 209-284. arXiv:math.AG/0208144.

49. Goncharov, A. "Polylogarithms, regulators and Arakelov motivic complexes" J. Amer. Math. Soc. 18 (2005), no. 1, 1–60.
50. Deligne P., Goncharov, A. "Groupes fondamentaux motiviques de Tate mixte" Ann. Sci. cole Norm. Sup. (4) 38 (2005), no. 1, 1–56.
51. Goncharov A. "Regulators" Algebraic K-theory handbook. Springer (2005), vol 1, 295-351. ArXiv: math.NT/0407308
52. Fock V., Goncharov A. "Moduli spaces of local systems and higher Teichmuller theory". Publications Mathematiques de l'IHES. v. 103 (2006) 1-211. ArXiv: math.AG/0311149.
53. Fock V., Goncharov A. "Cluster X-varieties, amalgamation and Poisson-Lie groups". Algebraic Geometry and Number Theory, In honor of Vladimir Drinfeld's 50th birthday, Birkhauser, (2006) Boston. 27-68. ArXiv: math.RT/0508408.
54. Gangl H., Goncharov A., Levin A.. "Multiple logarithms and trees". In: "Frontiers in Number Theory, Physics and Geometry" II: 759–774, Springer, Berlin, 2007.
55. Fock V., Goncharov A. "Convex projective structures on surfaces" Advances in Math. vol 208 (2007) 249-273. ArXiv: math.DG/0405348.
56. Fock V., Goncharov A. "Dual Techmüller and lamination spaces". Handbook of Teichmuller theory. Volume 1. IRMA Lectures in Mathematics and Theoretical Physics 11. European Mathematical Society. 647-684. math.DG/0510312.
57. Goncharov A.B., "Pentagon relation for the quantum dilogarithm and quantized $\mathcal{M}_{0,5}$ ". Progress in Mathematics, Vol 256, 413-426. (2007) Birkhauser Verlag Basel/Switzerland, arXiv:0706.4054.
58. Fock V., Goncharov A.B. "Cluster ensembles, quantization and the dilogarithm II. The intertwiner" Special volume dedicated to Yu. I. Manin's 70th birthday. Birkhauser Verlag Basel/Switzerland, (2008) 559-678, arXiv:math/0702398.
59. Goncharov A. "Euler complexes and geometry of modular varieties". Geometric Analysis and Functional Analysis. Geometry and Functional Analysis. Vol 12, (2007) 1873-1914. ArXiv: math.NT/0510310.
60. Gangl H., Goncharov A., Levin A.. "Multiple polylogarithms, polygons, trees and algebraic cycles". Proceedings of the Seattle conference on Algebraic geometry. Proceedings of Symposia in Pure Mathematics. Volume 80, Part 2, 547-594.
61. Fock V., Goncharov, A. "The quantum dilogarithm and representations of quantum cluster varieties". Inventiones Mathematicae. 175, 223-286 (2009).
62. Fock V., Goncharov A. "Cluster ensembles, quantization and the dilogarithm" Annales Scientifiques L'Ecole Normal Superier, t. 42, 2009, 865-930.
63. Goncharov A. "Hodge Correlators II". Moscow Mathematics Journal. Special issue dedicated to P. Deligne. 2010. 49 pages. arXiv:0807.4855.
64. "Classical Polylogarithms for Amplitudes and Wilson Loops" Alexander B. Goncharov, Marcus Spradlin, C. Vergu, Anastasia Volovich: 11 pages, Phys.Rev.Lett.105:151605, 2010, arXiv:1006.5703

65. Fock V.V., Goncharov A.B.: "Cluster Poisson varieties at infinity". *Selecta mathematicae*. Volume 22, Issue 4, (2016), 2569-2589. arXiv:1104.0407.
66. Goncharov, A.B., Kenyon R.: "Dimers and cluster integrable systems". *Annales Scientifiques L'Ecole Normal Superier*, 2013, arXiv:1107.5588. 78 pages.
67. Goncharov A.B.: "A simple construction of Grassmannian polylogarithms". arXiv:0908.2238. 26 pages, *Advances in Mathematics*, April 2013.
68. Golden J., Goncharov A.B., Spradlin M., Vergu C., Volovich A.: *Motivic Amplitudes and Cluster Coordinates*. 61 pages, arXiv:1305.1617 *Journal of High Energy Physics (JHEP)* 1401 (2014) 091.
69. Goncharov A.B.: "Hodge correlators" *Journal für die reine und angewandte Mathematik (Crelle's journal)*. (2019) vol 748, pp1-138.
70. Arkani-Hamed N., Bourjaily J., Cachazo F., Goncharov A.B., Postnikov A., Trnka J.: "Grassmannian Geometry of Scattering Amplitudes" arXiv:1212.5605, *Monographs in mathematical physics*. Cambridge University Press, 2016. 191 pages.
71. Goncharov A.B., Shen L.: "Geometry of canonical bases and mirror symmetry" 146 pages. arXiv:1309.5922 *Inventiones math.* (2015) v. 202, pp. 487-633.
72. V.V. Fock, A.B. Goncharov: "Symplectic double for moduli spaces of G-local systems on surfaces" *Advances in Mathematics*, September 2016, p. 505-543. arXiv:1410.3526.
73. Goncharov A.B.: "Exponential complexes, period morphisms, and characteristic classes." *Annales de la Faculte des Sciences de Toulouse*, Vol XXV, n 2-3, (2016) pp 397-459. ArXiv:1510:07270.
74. Dimofte T., Gabella M., Goncharov A. B.: "K-Decompositions and 3d Gauge Theories", *Journal of High Energy Physics*, 2016(11), 1-147.
75. Goncharov A.B.: "Ideal webs, moduli spaces of local systems, and 3d Calabi-Yau categories", in "Algebra, Geometry, and Physics in the 21st Century". Kontsevich Festschrift. *Progress in Mathematics*, 324, 2017, p. 31-99.
76. Goncharov A.B., Shen L.: "Donaldson-Thomas transformations for moduli spaces of G-local systems". *Advances in Math.* 2018, Volume 327, p. 225-348, arXiv:1602.06479.
77. Goncharov A.B., Zhu G.: "The Galois group of the category of mixed Hodge-Tate structures." *Selecta Mathematica*, 2018, 24(1), p. 303-358, Special volume dedicated to A.A. Beilinson. arXiv:1705.08175.
78. Finkelberg M., Gaitsgory D., Goncharov A., Polishchuk A.: "A tribute to Sasha Beilinson". *Selecta Mathematica*, 24(1), 1-5 . Published online 2-16-2018.
79. Bezrukavnikov R., Finkelberg M., Gaitsgory D., Goncharov A., Varshavsky Y.: "A tribute to David Kazhdan". *Advances in Math*, 2018, Volume 327, Pages 1-3.
80. Goncharov A.B., Shen L.: "Quantum geometry of moduli spaces of local systems and Representation Theory". arXiv 1904.10491, 235 pages. Accepted for publication, *Asterisque*, 2025.

81. Goncharov A.B., Rudenko D.: "Motivic correlators, cluster varieties and Zagier's conjecture on $\zeta_F(4)$ ". 106 pages. arXiv:1803.08585. Submitted.
 82. Goncharov A.B.: "Motivic fundamental groups of $G_m - \mu_N$ and geometry of modular manifolds". arXiv 1910.10321, 72 pages. Submitted.
 83. Goncharov A.B., Kislinsky O.: "Cluster construction of the second motivic Chern class". *Selecta Mathematica*, 29, 2023, N4, 45 pages.
 84. George T., Goncharov A.B., Kenyon R.: "The inverse spectral map for dimers " *Mathematical Physics Analysis and Geometry*, 2023. Published online September 16, 2023. 46 pages.
 85. Goncharov A.B., Kontsevich M.: "Spectral description of non-commutative local systems on surfaces and non-commutative cluster varieties". *Arithmetic and Algebraic Geometry, A Mathematical Tribute to Yuri Manin*. Simons Simposia, Yuri Tschinkel editor, Springer, 2024. 134 pages.
 86. Goncharov A.B., Sun Z.: "Exponential volumes of moduli spaces of hyperbolic surfaces". arXiv 2411.01615, 65 pages. Submitted
 87. Goncharov A.B.: "Archimedean hight pairings and generalized cross-ratios". *Proceedings of the conference Regulators V*, Pisa, 2024. To appear.
 88. Goncharov A.B.: "Quantum polylogarithms". 29 pages, 2025, Submitted, posted in the ArXiv 2601.00472.
 89. Goncharov A.B., Kontsevich M.: "Non-commutative cluster Lagrangians". 49 pages. 2026, ArXiv 2601.07538.
- Preprints available in the ArXive.*
90. Goncharov, A.B.: "Hidden Hodge symmetries and Hodge correlators". 9 pages Arbeitstagung. MPI Bonn June 2011. arXiv:1107.5710.
 91. Goncharov, A. "Periods and mixed Tate motives" arXiv:math.AG/0202154, 103 pages.
 92. Goncharov, A. "Multiple polylogarithms and mixed Tate motives" arXiv:math/0103059, 87 pages.